Introduction to Mutimodal AI for Biomedical Applications

[W1-1: Course Orientation]

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Department of AI

Course Information

- Course number: 59184
- Location: Building 310, Room 622
- Time: Tuesday 3, Thrusday 3, 4

Course Objectives

- Types and characteristics of multimodal data.
- History and fundamental principles of neural network-based artificial intelligence.
- Neural network architectures for processing different modalities.
- Multimodal AI techniques and their applications in the biomedical domain.

Da Vinci Learning (다빈치러닝) Format

Tuesday — Online Lecture

Watch the lecture video uploaded to e-class.

Thursday — Offline Activity

Participate in an offline class activity based on Tuesday’s lecture.

Da Vinci Learning (다빈치러닝) Format

Tuesday — Online Lecture

Watch the lecture video uploaded to e-class.

Focusing on understanding the key concepts and terminology.

Da Vinci Learning (다빈치러닝) Format

Thursday — Offline Activity

Participate in an offline class activity based on Tuesday's lecture.

Focusing on applying and reinforcing the concepts learned in the lecture.

Da Vinci Learning (다빈치러닝) Format

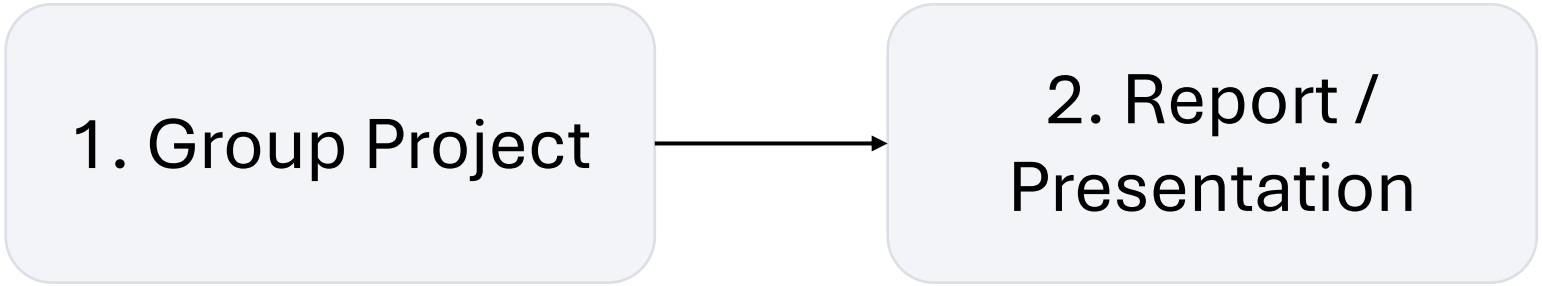
Thursday — Offline Activity

1. Individual Work → 2. Group Work → 3. Report

> **Without AI:** Individual Work → Group Work → **Compare with AI** → Report
>
> **With AI:** Individual Work → Group Work → Report

Da Vinci Learning (다빈치러닝) Format

Thursday — Offline Activity



Group Assignment

- Will be assigned after the course add/drop period.
- Students will be randomly assigned to groups while ensuring a balanced mix of students from different academic years.
- Assignment result will be announced on e-class.

Course Roadmap

Tuesday (Online)	Thursday (Offline)
• Week 01: Orientation	Introduction to AI
• Week 02: Introduction to Neural Network	Detailed instructions for each activity will be posted on e-class.
• Week 03: Neural Network – Architecture	
• Week 04: Neural Network – Training (1)	
• Week 05: Neural Network – Training (2)	
• Week 06: Neural Network – Training (3)	
• Week 07: Neural Network – Evaluation	
• Week 08: Midterm Exam	

Course Roadmap

Tuesday (Online)	Thursday (Offline)
• Week 09: • Week 10: • Week 11: • Week 12: • Week 13: • Week 14: • Week 15:	The second half of the course will cover topics such as CNNs, RNNs, and Transformers. Detailed schedule will be announced later.

• Week 16: Final Exam

Evaluation

- Attendance 5%
- Activity 15%
- Mid-term Exam 40%
- Final Exam 40%

Electronic Attendance

- If you forget to check in, attendance will not be granted afterward, even if you can prove that you attended the class.
- Please remember to check in through the electronic attendance system during every class.

Introduction to **M**utimodal **AI** for **B**iomedical **A**pplications

Welcome to **MABA**!